
Chia-An (Dennis) Hou

Phone: (+886)912-910-923, Email: chia.an.hou@gmail.com,

LinkedIn: [linkedin.com/in/chia-an-hou-ba178b274](https://www.linkedin.com/in/chia-an-hou-ba178b274)

ResearchGate: https://www.researchgate.net/profile/Chia-An-Hou-2?ev=hdr_xprf

Education

National Taiwan University (NTU) Taipei, Taiwan

Master of Science in Psychology

02/2026 – Present

Bachelor of Science in Psychology

09/2021 – 06/2025

GPA: 4.19 / 4.30, Rank: 2 / 74 (Top 3%)

Research Experience

National Taiwan University, Modeling and Informatics Lab (MILab), Taipei, Taiwan

Advisor: Tsung-Ren Huang, Ph.D.

Graduate Researcher

09/2025–Present

- Develop computational models utilizing Social Robots and LIWC to detect early-stage dementia risks through natural language processing (NLP) of daily conversations.
- Apply advanced statistical techniques in **R** and **Python** to analyze complex behavioral datasets and refine predictive accuracy.

National Taiwan University, Explorer of Perception and Attention (EPA) Lab, Taipei, Taiwan

Advisor: Su-Ling Yeh, Ph.D.

Undergraduate Research Assistant

01/2025–07/2025

Project: Industry-Academia Collaboration with Delta Electronics (Chatbot Emotion Regulation)

- Designed and executed HCI experiments to evaluate user experience and emotional resonance within **LLM-based chat interfaces**, focusing on the impact of linguistic tone and persona.
- Analyzed behavioral and physiological data using **R** to optimize interface efficiency, user engagement, and aesthetic preference.
- Collaborated with cross-functional engineering teams at **Delta Electronics** to translate psychological findings into actionable UI/UX design specifications.

National Taiwan University, Center for Artificial Intelligence and Advanced Robotics,
Taipei, Taiwan

Undergraduate Research Assistant

02/2023–07/2025

Project: AI Social Robot System Project

- Developed and optimized interaction personas and linguistic tones for social robots to enhance human-robot rapport.
- Spearheaded the technical implementation of chatbot frameworks and front-end web interfaces, incorporating **LLM API integrations** for real-time robotic responses.
- Conducted large-scale data collection for user engagement studies, ensuring rigorous data integrity and ethical compliance.

Publications & Manuscripts

Journal Manuscripts (In Preparation & Under Review)

Hou, C. A., Kuah, C. Y., Jhou, J. T., Yeh, S. L., Chang, Y. L., Chen, Y. C., Goh, J., Fu, L. C., & Huang, T. R. (Under Review). *From Upstream Semantic Association Deficits to Downstream Lexical Simplification in Mild Cognitive Impairment*. Submitted to **Journal of Medical Internet Research (JMIR)**.

Hou, C. A., Yeh, S. L., Chang, Y. L., Chen, Y. C., Goh, J., Fu, L. C., & Huang, T. R. (In Progress). *Mechanisms of Verbal Offense in Human-Computer Interaction Scenarios*.

Conference Presentations & Posters

Zheng, P. Y., **Hou, C. A.**, Pan, C. H., Liu, Y. C., Liu, T., Chiu, T. W., & Yeh, S. L. (Under Review). Can Talking to AI Chatbots Help When You're Feeling Down? Examining the Role of Perceived Empathy. Submitted to **2026 IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)**.

Hou, C. A., Kuah, C. Y., Jhou, J. T., Chen, Y. C., Chang, Y. L., Fu, L. C., Yeh, S. L., & Huang, T. R. (2026, August 6–8). *Sensing Reduced Lexical Complexity as a Behavioral Marker of Mild Cognitive Impairment through Social Companion Robots*. Poster presented at the **American Psychological Association (APA) 2026 Convention**, Washington, D.C., USA.

Chen, Y. C., & **Hou, C. A.** (2025). *Mema!d: Designing a Gamified Cognitive Assessment and Family Sharing Mechanism for Senior Health Promotion*. Poster presented at the **TAIwan Computer-Human Interaction Conference (TAICHI'25)**, Taipei, Taiwan. [Presented in Mandarin]

Hou, C. A., Chen, Y., Yang, Y., Huang, T., Wang, Y., & Wang, E. (2024, October). *Behind the Vote: Assessing AI's Predictive Power in American Elections*. Oral presentation at the **63rd Annual Convention of the Taiwanese Psychological Association**, Taipei, Taiwan.

Hou, C. A., Lin, Y., Pan, T., Hong, R., & Chiang, C. (2024, July). *Examining the Impact of Camera and Avatar Usage on Video Conferencing Experiences: A Study of Taiwanese College Students*. Paper presented at the **TAIWAN Computer-Human Interaction Conference (TAICHI'24)**, Tainan, Taiwan.

Honors & Awards

- **Dean's Award**, College of Science, NTU 05/2025
 - *Awarded to the top 10% of graduating students for outstanding academic performance.*
- **Dr. Yu-Chun Chung Memorial Scholarship** 05/2025
 - *A merit-based scholarship awarded to psychology students with excellent academic records and research potential.*
- **Altruism Award**, College of Science, NTU 05/2024
 - *Recognized for exceptional service to the community and dedication to the welfare of others.*
- **NSTC Undergraduate Research Grant** 02/2024
 - *A prestigious national grant awarded to undergraduates with high-potential research proposals (Acceptance rate approx. 10-15%).*
- **Presidential Award**, NTU Spring 2022, Fall 2023, Spring 2024
 - *Awarded to the top 5% of students in each department based on semester GPA.*

Teaching & Mentorship

National Taiwan University, Department of Psychology, Taipei, Taiwan

Teaching Assistant 02/2026 – Present

Course: Neural and Behavioral Modeling
(Instructor: Tsung-Ren Huang, Ph.D.)

- Facilitated weekly laboratory sessions, guiding students through statistical modeling and experimental programming in **R**.
- Evaluated assignments and provided technical support for student research projects on computational cognitive science.

National Taiwan University, D-School, Taipei, Taiwan

Teaching Assistant 02/2024 – 06/2024

Course: Reset Your Mind: Finding Your Way in College
(Instructor: Ping-Cheng "Benson" Yeh, Ph.D.)

- Assisted in course orchestration and student mentorship for a high-impact course focused on growth mindset and design thinking.
- Facilitated group discussions and managed peer-review processes to enhance student engagement and reflective learning.

National Taiwan University, Harvard Taipei Academy, Taipei, Taiwan

Student Volunteer

07/2024 – 08/2024

- Served as a cultural ambassador and peer mentor for Harvard University students during their intensive language and culture program in Taiwan.
- Coordinated cross-cultural exchange activities and provided linguistic support, fostering international academic collaboration.

Technical Skills

- **Programming:**
 - **Proficient:** Python (Data Analysis, Visualization, Deep Learning, LLM API integration); R (Statistical Modeling, ggplot2).
 - **Fundamental:** JavaScript, HTML/CSS (Front-end Development); SQL (Database Management); STATA.
- **Research Tools:** LaTeX, Git/GitHub, SPSS, PsychoPy/Qualtrics (Experimental Design).
- **Languages:** Mandarin (Native); English

Leadership & Practical Experience

Left Bank Project

Project Leader

02/2026 – Present

- **Awarded the NTU DLC Self-Directed Learning Grant.**
- Collaborated with NTU Psychology Department & Alumni Association
- Curated a series of community-building events and career development workshops, fostering interdisciplinary dialogue among students, faculty, and alumni.

AI Psychology Podcast “Prompt Your Life”

Co-Founder & Podcast Host

02/2026 – Present

- **Awarded the NTU DLC Self-Directed Learning Grant** for innovative psychological knowledge translation.
- Launched and co-hosted a podcast focused on applying psychological principles and AI tools to daily life challenges.